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T4 _____	A	F4 _____

2018
HiMCM
Summary Sheet

(Your team's summary should be included as the first page of your electronic submission.)
Type a summary of your results on this page. Do not include the name of your school, advisor, or team members on this page.

Summary

There are various kinds of roller coaster around the world, which appeal to many fans. In order to recommend special roller coasters to fans, some online ranking of roller coasters come out in different website. Those ranking are based on either specific objective indicator or subjective experience results. therefore, we should create a quantitative algorithm with multiple objective indicators and build a new ranking of roller coaster, compare our ranking with two other online ranking systems. In addition, as the APPs have embedded in our lives, so we also should design a friendly APP which can recommend different roller coaster to fans based on their preferences.

In the database of COMAP_RollerCoasterData_2018.xlsx, we find that a small amount of data doesn't fit the facts and some indicators have more missing data, so we should process the data. Firstly, we correct the wrong data, secondly, we divide the sample into different categories by cluster analysis, and then we made up for the missing data by the mean value of those categories.

For creating a quantitative algorithm, we should set several assumptions. The roller coaster is operating normally, the ticket price of the roller coaster and the weather does not affect our ranking, the height, speed, length, the number of inversions, duration, G force and vertical angle are directly proportion to the fans' experience. We rank the roller coaster with entropy weight method. According to the character of the indicator in the database, we construct five indexes, including ups and downs index, vertigo index, sustainability index, stimulation index and nostalgia index. After our calculation, Steel Dragon 2000, Smiler, Kinda Ka, Leviathan, Fury 325, Millennium Force, Top Thrill Dragster, Intimidator305, Fujiyama and Steel Vengeance rank top 10, Our ranking are partly identical to the other online ranking, and finally, we cut the last 100 roller coasters in our ranking, and calculate the ranking again, we find the result is very identical with our original ranking, which means our result is robustness.

According to our classification and ranking, we design a friendly APP to recommend the roller coaster. We classify five indexes into excitement, intensity and vertigo, and then we take the fans' preference into account. The fans can input his preference into our APP, he will find the roller coaster which rank the top.

We have three advantages, including the correction of error data and supplement of missing data, entropy weight method based on the preference, and friendly app design. Surely, a coin has two sides, our disadvantages include the supplement of missing data leading to the results bias, and the results depending on our preference.

Keyword: Cluster Analysis, Entropy Weight Method, Personalized Comprehensive Evaluation Model

Rank it! Go to it! Ride it!

Have you been feeling the blood pumping in your heart, eager to find a roller coaster to let epinephrine wake you up? Have you had night that you couldn't sleep, thinking about flying in the sky, hearing the screaming howling? Or have you been disappointed by those dull, low-quality roller coasters, wanting to say good-bye to the most exciting game in this world? Forget about those subjective assessed roller coaster ranking websites. Rank&Roll, a newly published app with the latest algorithm, will provide you with the most objective ranking in the world!

To meet the personal preferences of the users, Duncan's team has designed an application to help each individual to get personal roller coaster ranking list. When first signed up, clients will complete the following processes in order to provided information to sort out the most suitable list.

They have already given out a top 10 list of roller coasters, as shown below:

Name	Score
Steel Dragon 2000	5.05
Smiler	4.00
Kingda Ka	3.88
Leviathan	3.70
Fury 325	3.68
Millennium Force	3.50
Top Thrill Dragster	3.26
Intimidator 305	3.25
Fujiyama	3.05
Steel Vengeance	2.96

How well will this app perform? I expect it will be quite popular among roller coaster fans. Let's wait and see.

Get Ready for Scream

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1. Introduction

Technology is booming. After going through a boom in electronic devices, people gradually get used to it. Thus, groups of people return to theme parks, seeking thrills in places such as roller coasters. Roller coaster lovers naturally want to find the most appropriate roller coaster. Because there is no objective ranking, they can only be based on the Internet-given rankings. However, the online ranking is usually given by individual evaluators according to their personal feelings or it's just formed by a bunch of subjective votes.

In order to solve this problem, our group decided to use a series of algorithms to obtain the ranking based on the main objective data of roller coaster. Then to justify the objectivity of the rank we have worked out, we will compare and discuss the rating results and descriptions from your team's algorithm with two other ranking systems found online. We will design and develop the concept of a user-friendly application in order to meet the personal preference of each individual. The application will base on our previous algorithm and apply screening system thus provide a personal roller coaster ranking. At last, our team will write a one-page non-technical News Release describing our new algorithm, results, and app.

When making the ranking of roller coasters, we have to fill in the information first. We decide to use cluster analysis to use the data of similar roller coasters to approximate the information. After the completion of information, we used entropy method with two layers of indexes to make the ranking list. To design the app, we also used the two-layer structure to rank the roller coasters. The options are designed in this way to enhance users' experience. We also used the collaborative filtering to give recommendation to old users, whose long-term preference has been saved by the app. To achieve the fixed amount of roller coasters to recommend, we used k-nearest-neighbor method to get the recommended roller coasters.

2. Assumptions and Justifications

Assumption 1: No accident will happen when the roller coasters are operating. That is to say, all the roller coasters are absolutely safe, so safety factors will not be considered in our ranking system.

Justification: The possibility that a roller coaster has an accident is approximately one in 250 million, which is so small. Plus, safety factor is not included in the database, so we have no way to take them into consideration.

Assumption 2: Our ranking system do not consider the price of the tickets for the roller coasters.

Justification: In our ranking system, only factors related riders' experience will be taken into consideration.

Assumption 3: Weather factors is not included in the ranking system.

Justification: As roller coasters are everywhere around the world, the climate at their locations may vary drastically, which might also affect the riding experience. However, it is extremely hard for us to collect so much data in such a short period of time. So, we decide not to consider the climate in the ranking system.

Assumption 4: A rise in height, speed, length, the number of inversions, duration, G force and vertical angle will all increase the riders' excitement without causing any unpleasantness.

Justification: In the Entropy Method we use, all the variables and final score is positively correlated, because we assume that people ride the roller coasters to look for excitement.

3 Index Needed

Table 1 Index Needed

Index	Meaning of the indexes
s_i	ups and downs index of the *ith* roller coaster， $i=1,\cdots,300.$
n_i	vertigo index of the *ith* roller coaster
h_i	sustainability index of the *ith* roller coaster
t_i	stimulation index of the *ith* roller coaster
Y_i	nostalgia index of the *ith* roller coaster
a_j	preference coefficient of the *jth* index， $j=1,\cdots,5.$

4. Data Analysis and Processing

4.1 Screening and Revision of Erroneous Data

By classifying and screening the data in COMAP_RollerCoasterData_2018, there are eight types, among which wood and steel are the same as construction. There are three roller coasters involved. After checking on the Internet, the error data can be modified as shown in Table 2. In the process of consulting the data, we find that there are differences in very few numerical data. Since it is impossible to verify all the data, we use COMAP_RollerCoasterData_2018 in the later calculation.

Table 2 Correction of Erroneous Type Data

Name	Original Type	After modification
Terminator Salvation	Wood	Sit down
Cedar Creek Mine Ride	Steel	Sit down
Corkscrew（1976）	Steel	Sit down

4.2 Estimation of Missing Data for a Few Speed and Length

Through the analysis of the data, we find that there are a lot of missing data are drop, duration, G force and vertical angle data, and only a few speed and length data are missing. For the missing data of speed and length, we use the "approximate complement" method to estimate.

Let's assume that the speed of roller coaster A is missing. Choose the speed data of roller coaster B, whose height and length data is closest to roller coaster A, with the same type and

construction of roller coaster A as the estimated value of speed data of A. Set h_0, l_0 for value of the height and length of roller coaster A and h_m, l_m for data of all roller coasters with the same types and construction of roller coaster A. The roller coaster which meets the following formula

$$\min_k \sqrt{\left(\frac{h_k - h_0}{h_0}\right)^2 + \left(\frac{l_k - l_0}{l_0}\right)^2} \tag{1}$$

is B, and its corresponding speed value is v_B. The estimated value of speed for A is $v_0 = v_B$.

Similarly, missing length data can be estimated. If the length data l_B of roller coaster meets the following formula,

$$\min_k \sqrt{\left(\frac{h_k - h_0}{h_0}\right)^2 + \left(\frac{v_k - v_0}{v_0}\right)^2} \tag{2}$$

the missing data can be estimated as $l_0 = l_B$.

4.3 Estimation of Missing Data for Drop, Duration, G force and Vertical angle

Wooden roller coaster is often built along the hillside with rarely inversion. By analyzing the data of fifty groups of wooden roller coasters in COMAP_RollerCoasterData_2018, there are only 5 of them have inversions, and the number of inversions is not more than 3. Because people always think that wood is not as strong as steel in their subjective impression, the stimulation of insecurity of wooden roller coaster in riding is unmatched by steel. At the same time, because of the construction along the hillside, the feeling of ups and downs is much more intense than that of steel, when people ride a wooden roller coaster. However, the dizziness caused by inversion and rotation of wood is not comparable to that of steel. Therefore, the gap between these two kinds of construction roller coaster is still very obvious.

4.3.1 Estimation of Missing Data for Roller Coaster Made of Wood

Although the construction process of roller coaster is developing over time, the physical principles are the same. Therefore, ignoring the construction time factor, we selected 50 roller coasters with construction as wood, and carry out cluster analysis of roller coasters with height, speed and length as basic features. Fifty roller coasters' clustering maps are obtained as shown in Figure 1. We believe that the size and structure of roller coasters in the same category are similar, and their characteristic attributes should also be similar

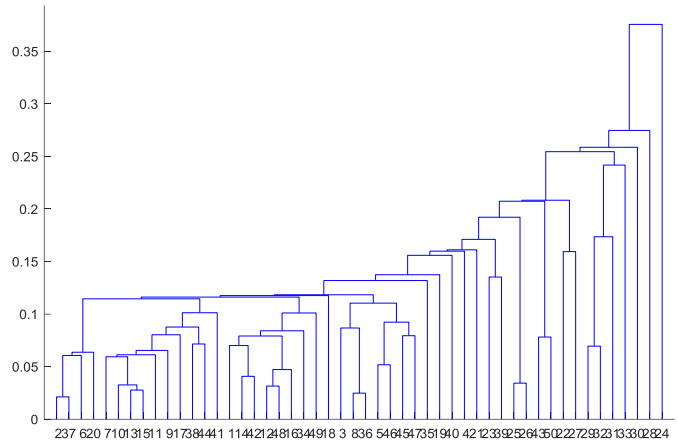


Figure 1 Cluster Graph of 50 Wood Roller Coasters

Of the 50 roller coasters, 18 are missing drop data, and the number of roller coasters missing data is less than the number of those not missing data. For roller coaster A with missing data, select in category with A the most similar roller coaster B with drop data, and use drop data of B to estimate that of A. If there are more than one such B, the average value of multiple drop data is used to estimate. The more the number of roller coasters not missing data is, the more detailed the clustering can be, and the closer the estimated value is to the real value.

For duration data, there are 14 roller coasters missing data. The situation is similar to that of drop data, and the data supplement method is the same as drop data.

There are 28 roller coasters missing data of vertical angle data, and the number of roller coasters with missing data is more than the number of those without missing data. It may appear that the roller coasters similar to the A in category are all missing vertical angle data. Therefore, we can consider to increase the number of roller coasters of the same kind as A by rough classification, so as to make roller coaster B without missing data as possible, which enables vertical angle data to be estimated. But this may increase the estimation error.

Only 7 of the 50 roller coasters have G force data, so it doesn't work to estimate with known data of the same category after cluster analysis. Because the 7 data may be a cluster, we should drop it. If the data is missing, it can be estimated according to the known data. Otherwise, other methods shall be used. By consulting relevant information on the internet, the G force of roller coaster will not exceed 2.5G without considering the car's own acceleration system. Because G force is an important indicator of the performance of roller coaster, it should be open data. If there is no relevant data or relevant data is difficult to find, it is reasonable that the roller coaster carriage does not have an acceleration system. In that case, its G force will not exceed 2.5G, and we might as well assume that it is 2G.

4.3.2 Estimation of Missing Data for Roller Coaster Made of Steel

The same method used for roller coasters made of wood can be used when estimating the missing data of roller coasters made of steel. We find from the data that 136 of 250 roller coasters made of steel have inversions and 114 not, so we should divide them into two parts according to whether they have inversions or not. For each part, the missing data can be estimated according to the method in Section 4.3.1.

When the roller coaster is built without inversion, we use height, speed, length as the three basic features for cluster analysis, while we use height, speed, length, number of

Inversion as the four basic features for cluster analysis, when it is built with inversion. The two-part cluster analysis schematic diagram is shown in Figure 2 and Figure 3, in which we only show the clustering results of the first 30 roller coasters in each part.

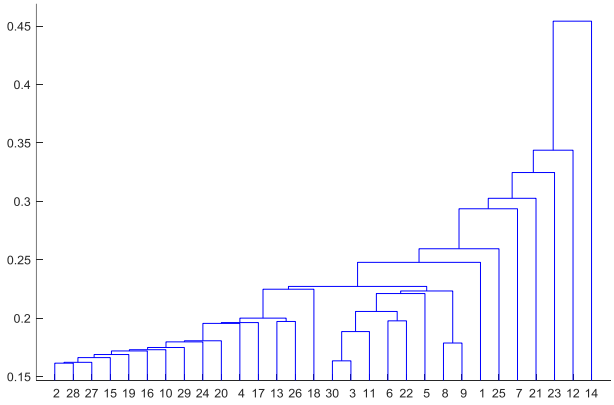


Figure 2 Cluster Diagram with Inversion

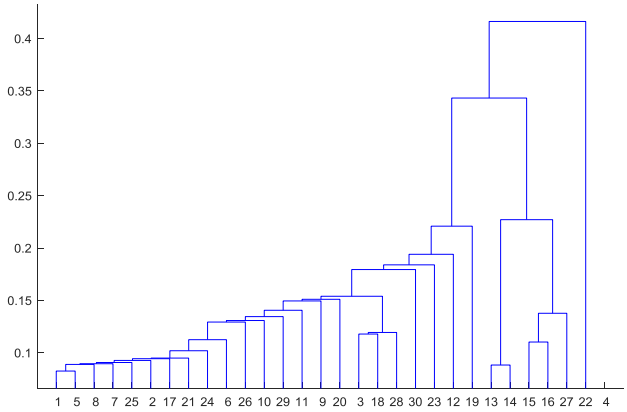


Figure 3 Cluster Diagram without Inversion

5. Evaluation Based on Entropy Weight Method

5.1 Analysis and Quantification of Characteristics

Through the method of section 4, we have revised and supplemented the data. For 300 roller coasters, each evaluation object contains 12 characteristics: type, construction, Year/Date, speed, length, inversions, number of inversions, drop, duration, G force and vertical angle. By consulting the relevant literature, passengers' ride experience varies when there are reversions or not. Roller coaster without inversions tend to use the alternating transformation of overweight and weightlessness caused by rapid changes of acceleration as the main stimulation. While those with inversions tend to use inversions and the vertigo caused by rotation as the main stimulation. The inversions characteristic can be ‘yes’ quantized to 1, and the ‘no’ quantification is 0.

Through the information online and the analysis of data, roller coasters made of wood rarely have inversions, while steel-made roller coaster does not. By comparing 50 wood-made roller coasters and 114 steel-made roller coasters which don’t have inversions, The average data of height, speed and length are obtained in Table 3.

Table 3 Comparison of Mean Values Between the Both

Construction	Height（feet）	Speed（mph）	Length（feet）
Wood	110.22	56.045	3726.77
Steel	140.48	59.90	3040.80

Basically, these data can be found on the Internet, not estimated. As you can see from table 3, those steel-made roller coasters without inversions are higher and faster than wood-made ones, thus the average degree of stimulation is higher. The length data of wood-made roller coaster is higher than steel-made, which means that the average duration of stimulation from wood-made roller coaster is longer. The degree of Steel-made ones higher than the wood-made can be depicted by the following formula:

$$\left(\frac{140.48-110.22}{110.22}+\frac{59.9-56.045}{56.045}\right)\Big/2=0.172 \tag{3}$$

So when we quantify 'steel' to 1, the quantization value of 'wood' is 0.828.

According to the information on the internet, the sensation intensity of human body is different when choosing different types of roller coasters, even under the same level of stimulation. This is because the ability to control our body when experiencing different types of roller coasters varies a lot. According to the intensity of sensory strength from weak to strong, the arrangement and quantitative rules of the 6 types can be seen in Table 4.

Table 4 Quantization of These 6 Types

Type	Sit down	Stand up	Wing	Flying	Suspended	Inverted
Quantization	1	2	3	4	5	6

5.2 Evaluation Feature Extraction

5.2.1 Ups and Downs Index

According to the analysis of 5.1, we know that when roller coaster does not have inversions, the stimulation mode for passengers is mainly the alternating transformation of overweight and weightlessness caused by rapid change of acceleration. Therefore, we define the index of ups and downs, which can be characterized by characteristic speed, drop, G force and vertical angle. These four characteristics are positively correlated with this index.

Entropy represented, as we know, the overall degree of disorder of the whole system. The larger entropy is, the more information it would carry, which means a larger degree of disorder. Thus, a large entropy would mean a greater uncertainty of an index. Thus, a smaller entropy would mean a greater accuracy and contribution to the whole model. With this method, we can determine the factors that contribute most to the assessment equation, allowing us to work out the weight of each factor. At last, we can use the equation to give an overall assessment of the roller coasters.

When the system may be in several different states and the probability of each state is p_i（$i=1,2,\cdots,m$）, the entropy of the system can be defined：

$$e=-\frac{1}{\ln m}\sum_{i=1}^{m}p_i\ln p_i \tag{4}$$

Entropy weight method is an objective weighting method. In the specific use process, the entropy weight method calculates the entropy weight of each feature according to the degree of variation of each feature, and then amends the weight of each feature through the entropy

weight, thus objectively obtaining the weight of each feature. We set $j=1,\cdots,4$ corresponding to characteristics: speed, drop, G force and vertical angle. p_{ij} represents the proportion of the *ith* roller coaster about the *jth* characteristic:

$$p_{ij}=\frac{a_{ij}}{\sum_{i=1}^{300}a_{ij}},\ i=1,2,\cdots,300,\ j=1,\cdots,4 \tag{5}$$

Where, a_{ij} represents the original data of the *ith* roller coaster's *jth* characteristic.

Calculate the entropy of the *jth* characteristic:

$$e_j=-\frac{1}{\ln 300}\sum_{i=1}^{300}p_{ij}\ln p_{ij},\ j=1,\cdots,4 \tag{6}$$

Calculate coefficient of variation of the *jth* characteristic:

$$g_j=1-e_j,\ j=1,\cdots,4 \tag{7}$$

Thus the weight of the *jth* characteristic:

$$w_j=\frac{g_j}{\sum_{j=1}^{4}g_j},\ j=1,\cdots,4 \tag{8}$$

So we can get the *ith* roller coaster's index of ups and downs:

$$s_i=\sum_{j=1}^{4}w_j p_{ij},\ i=1,2,\cdots,300 \tag{9}$$

After calculation, the weight coefficient of speed, drop, G force and vertical angle is 0.1864,0.5489,0.1967,0.0680 respectively, therefore

$$s_i=0.1864p_{i1}+0.5489p_{i2}+0.1967p_{13}+0.0680p_{14},\ i=1,2,\cdots,300 \tag{10}$$

From the weight coefficient, we can see that drop feature plays the most important role in the ups and downs index.

5.2.2 Vertigo Index

Those roller coasters with inversions tend to use inversions and the vertigo caused by rotation as the main stimulation. Thus we define the index of vertigo which can be described by characteristics inversions and number of inversion. But these two characteristics have strong correlation. As long as number of inversions is not zero, inversions must be 1. Otherwise, as long as number of inversions is 0, inversions must be 0, so only inversions index can be adopted. We need to normalize the number of inversions. The vertigo index $\tilde{n}_i$ standardized from the *ith* roller coaster' number of inversions is calculated as follows:

$$\tilde{n}_i=\frac{n_i-\min\limits_i n_i}{\max\limits_i n_i-\min\limits_i n_i} \tag{11}$$

Where n_i is the number of inversions of the *ith* roller coaster.

5.2.3 Sustainability Index

People who like roller coaster often like to pursue stimulation, and an important aspect of pursuing stimulation is the duration of stimulation, which is closely related to the scale of roller coaster. Therefore, the index can be characterized by height, length and duration, and the sustainability index is in positive phase with the three characteristics.

As the calculation of the ups and downs index, the value of the *ith* roller coaster's sustainability index h_i can be obtained by using the Entropy Weight Method.

After calculation, the weight coefficient of height, length and duration is 0.0057,0.0032,0.9911 respectively, therefore

$$h_i = 0.0057\,p_{i1} + 0.0032\,p_{i2} + 0.9911\,p_{13},\ i = 1,2,\cdots,300 \tag{12}$$

From the weight coefficient, we can see that duration feature plays the most important role in the sustainability index.

5.2.4 Stimulation Index

Through the analysis of 5.1, we can see that when people ride roller coaster made of wood or steel, their subjective feelings are different. At the same time, when people choose different types of roller coasters, their subjective feelings of stimulus are different even when the other conditions are the same. Therefore, the stimulation index is depicted by characteristics construction and type.

According to the quantitative feature of type, the greater the value, the higher the corresponding stimulus sensitivity. In order to facilitate calculation, we need to normalize the type data. The type standardized data of roller coaster is calculated as follows:

$$\tilde{t}_i = \frac{t_i - \min\limits_i t_i}{\max\limits_i t_i - \min\limits_i t_i},\quad i = 1,\cdots,300. \tag{13}$$

Where t_i represents the type of the *ith* roller coaster. Set c_i as the construction data of the *ith* roller coaster .Through the Entropy Weight Method, we can get the *ith* roller coaster's stimulation index r_i.

5.2.5 Nostalgia Index

These 300 roller coasters are from 1924 to 2018. The time span is 94 years. The construction time of the *ith* roller coaster is indicated by y_i .Set $y_i' = 2018 - y_i$,and the nostalgia index is :

$$Y_i = \frac{y_i' - \min\limits_i y_i'}{\max\limits_i y_i' - \min\limits_i y_i'},\quad i = 1,\cdots,300. \tag{14}$$

This is a very personalized index.

5.3 Establishment and Solution of Evaluation Model

5.3.1 Comprehensive Evaluation Model Based on Passengers' Preference Degree

Section 5.2 proposes five evaluation indexes and the comprehensive evaluation here is based on these five aspects. We can design preference coefficient to match different preference levels and the comprehensive evaluation index of 300 roller coasters is:

$$f_i = a_1\tilde{s}_i + a_2\tilde{n}_i + a_3 h_i + a_4 r_i + a_5 Y_i,\ i = 1,\cdots,300. \tag{15}$$

Where coefficients $a_1,\cdots,a_5$ represent preferences for different indexes. $a_1,\cdots,a_5 \geq 0$ and $\sum_{i=1}^{5} a_i = 1$. The larger the value of a_i, the more important its corresponding index is in the evaluation system. In order to eliminate the influence of dimension, it is necessary to normalize the data before calculating the comprehensive evaluation index. $\tilde{s}_i$ is normalized from s_i:

$$\tilde{s}_i = \frac{s_i - \min\limits_i s_i}{\max\limits_i s_i - \min\limits_i s_i},\ i = 1, \cdots, 300. \tag{16}$$

When evaluating 300 roller coasters, the larger the value of f_i, the higher the corresponding roller coaster ranks. Based on the preference coefficient, we can make a personalized ranking scheme according to the different preferences of different passengers, which provides operational feasibility for us to design apps.

However, when we do not consider the personalized ranking scheme with preference degree and only hope to rank by characteristic data of roller coaster, we can still use the entropy weight method to determine the objective weighting coefficients of five indicators. So that the five indicators can be combined into a comprehensive evaluation index, ranking by calculating their comprehensive scores.

5.3.2 The Ranking Result and Comparison with Online Rankings

Using the entropy weight method to determine the weighting coefficient of the 5 indexes, we get a comprehensive evaluation index. After calculation, 300 roller coasters' comprehensive index scores are obtained, among which the top 10 roller coasters are listed in Table 5.

Table 5 Overall Ranking of Roller Coasters

Name	Score
Steel Dragon 2000	5.04881085
Smiler	4.00226545
Kingda Ka	3.88475056
Leviathan	3.70368266
Fury 325	3.68319054
Millennium Force	3.50044987
Top Thrill Dragster	3.26340141
Intimidator 305	3.25427553
Fujiyama	3.05363808
Steel Vengeance	2.96345454

5.3.3 Comparison with Online Ranking

We found two online roller coaster rankings and made a comparison with our ranking, as shown in Table 6.

Table 6 Result and Comparison with Online Rankings

Rank	Online1	Online2	Ours
1	Millennium Force	Kingda Ka	Steel Dragon 2000
2	Steel Vengeance	Dodonpa	Smiler
3	Top Thrill Dragster	X2	Kingda Ka
4	Maverick	Top Thrill Dragster	Leviathan
5	El Toro	Steel Dragon 2000	Fury 325
6	Fury 325	Tower of Terror	Millennium Force
7	Intimidator 305	Millennium Force	Top Thrill Dragster
8	The Voyage	Intimidator	Intimidator 305
9	Kingda Ka	Thunder Dolphin	Fujiyama
10	Apollo's Chariot	Eejanaika	Steel Vengeance

Note: two groups of online ranking data are derived from references.

After applying the algorithm presented in the previous text, we have made the following ranking.

To justify our ranking's reliability, we have searched several ranking systems online and have picked the two most authentic ranking to make comparison with our ranking.

The first ranking we will present is from the roller coaster system in "Ranker" community. Ranker as a ranking community, let people vote for everything, so the ranking on its web will be rather reliable and representative for the people who voted.

In the two rankings, there are 5 repetitions in the "top 10", which are Fury 325, Millennium Force, Kingda Ka, Intimidator 305 and Top Thrill Dragster. In our ranking, the order of those 5 from front to back is Kingda Ka, Fury 325, Millennium Force, Top Thrill Dragster and Intimidator 305. In another ranking, the sequence is Millennium Force, Top Thrill Dragster, Fury 325, Intimidator 305 and Kingda Ka.

The sequences of those two quite correspond except Kingda Ka.

Our second reference is from "THE TOP 10s". The same as "Ranker", "THE TOP 10s" is a ranking system which allows people to vote. As a result, this source is respectively reliable and Have reference value.

In the two rankings, there are 5 repetitions in the "top 10", which are Millennium Force, Kingda Ka, Intimidator 305, Steel Dragon 2000 and Top Thrill Dragster. In our ranking, the order of those 5 from front to back is Steel Dragon 2000, Kingda Ka, Millennium Force, Top Thrill Dragster and Intimidator 305. In another ranking, the sequence is Kingda Ka, top Thrill Dragster, Steel Dragon 2000, Millennium Force and intimidator 305.The sequences of those two quite correspond except Kingda Ka.

Moreover, as shown in our form, there are three repetitions which are Millennium Force, Top Thrill Dragster and Kingda Ka in the two ranking lists found online. This means that the repeated three roller coasters are confirmed to be eligible to be put in such high rankings. The three, furthermore, can all be found in our top lists, so a further confirmation that justify our objectivity is made.

Then comes the question: How can we explain the difference between the votes and our list? To answer this question, we have come up several possible explanations.

First, the ranking is based on the voters' personal experience. As we all know, different positions in the roller coaster can lead to different experiences. The more you sit back, the bigger the thrill. We have mentioned in our assumption that we would not consider the influence of positions of seats, so this might be a factor which lead the difference.

Moreover, the price of the tickets might also be an influential factor which may affect the votes. Sometimes a good roller coaster may not receive much votes for its high price. Likewise, we have mentioned that the price of the roller coaster will not make any influence in the assumption. This may also be a possible factor that causes the difference.

6. Sensitivity Analysis

In the above analysis, we have got the Roller coaster ranking by the comprehensive evaluation of multiple index with objective weight. For testing the robustness of our result, we use the new sample which drops the last 100 Roller coasters of our ranking, and then we calculate it again. the comparative result is shown in Table 7.

Table 7 The Outcome of Two Different Samples

300 Samples			200 Samples		
Name	Rank	Score	Name	Rank	Score
Steel Dragon 2000	1	5.05	Steel Dragon 2000	1	5.03
Smiler	2	4.00	Kingda Ka	2	4.03
Kingda ka	3	3.88	Smiler	3	3.96
Leviathan	4	3.70	Leviathan	4	3.76
Fury 325	5	3.68	Fury 325	5	3.69
Millennium Force	6	3.50	Millennium Force	6	3.51
Top Thrill Dragster	7	3.26	Top Thrill Dragster	7	3.40
Intimidator 305	8	3.05	Intimidator 305	8	3.32
Fujiyama	9	2.96	Fujiyama	9	3.05
Steel Vengeance	10	2.95	Coaster Through the Clouds	10	2.99

Through the comparison, we find that the ranking of two samples is rather identical, so our method is robust and convincing.

7. Design for the App

7.1 Explanation of the Ranking System in App

In Section 5, our evaluation algorithm fully considers the individual ranking requirements of roller coaster for different passengers because of their different preferences. To this end, we design the app to achieve personalized recommendation based on the preferences of passengers. Users only need to fill in a few related preferences, then we can personalize the sorting according to the algorithm and give the recommended roller coaster.

Considering that some passengers may have some very special preferences, for example, some passengers are paranoid about taking roller coaster made of wood, while others only take steel. The comprehensive evaluation model we have established is not focused on the distinction, so it can be listed as an option alone. It can be embodied in the following aspects.

1. Do you prefer wood to steel? There are three options -- wood, steel and both.

If the passenger chooses wood, then we will choose in the 50 roller coasters made of wood. If the passenger chooses steel, then we will choose in the 250 roller coasters made of steel. If passengers choose “both”, then we will choose in all 300 roller coaster.

2. Which type of roller coaster do you prefer? There are two options -- sit down and others.

This is because 240 of the 300 roller coasters are sit down, up to 80%. If passengers choose sit down, we will further select the roller coasters which meets the first option and whose type is sit down. If the passenger chooses others, we will choose in the other 60 roller coaster.

3. Do you have nostalgia? Do you prefer the old roller coaster to the modern one? There are three options -- ancient, modern and both.

If the passenger chooses the old one, his nostalgia degree is higher, and the parameter a_5 can take a larger value. Because this index is not an important index when evaluating roller coaster, its value should not more than 0.5. If the passenger chooses modern, the value of a_5

is smaller. If passengers choose "both", it means that the index has little impact on passengers, thus $a_5 = 0$.

4. Do you have any request for the ride time of the roller coaster? The three option is longer, shorter, and no demanding.

If the passenger chooses a longer time, then the passenger pursues the persistence in the stimulation process, and the value of the parameters a_3 can be larger; if the passenger chooses a shorter time, it can be smaller; if the passenger does not require it, then $a_3 = 0$. Although this index is important, its size should not larger than the parameters of ups and downs index and vertigo index.

5. Do you prefer fast alternating A for weightlessness and overweight caused by ups and downs, to B for vertigo caused by inversions and rotation? The three options are A, B and both.

If the passenger chooses "both", the parameter is $a_1 = a_2$, if the passenger chooses A, then there is $a_2 = 0$, if the passenger chooses B, then there is $a_1 = 0$.

After sorting out the selected roller coasters screened through 1 and 2, the evaluation model is

$$f_j = a_1\tilde{s}_j + a_2\tilde{n}_j + a_3 h_j + a_5 Y_j$$

Where the non-negative parameters a_1, a_2, a_3, a_5 may be 0 either, and only need to satisfy the condition that their sum is 1. j is the *jth* object of our roller coaster.

According to the calculation results, the individualized sorting of the selected roller coasters can be realized, and the best recommendation can be made to the passengers. For the simplicity of UI design interface and better customer self-evaluation, we classify ups and downs index, vertigo index, sustainability index, stimulation index and nostalgia index into tree feelings (excitement, intensity, vertigo), and the result of classification is in figure 4.

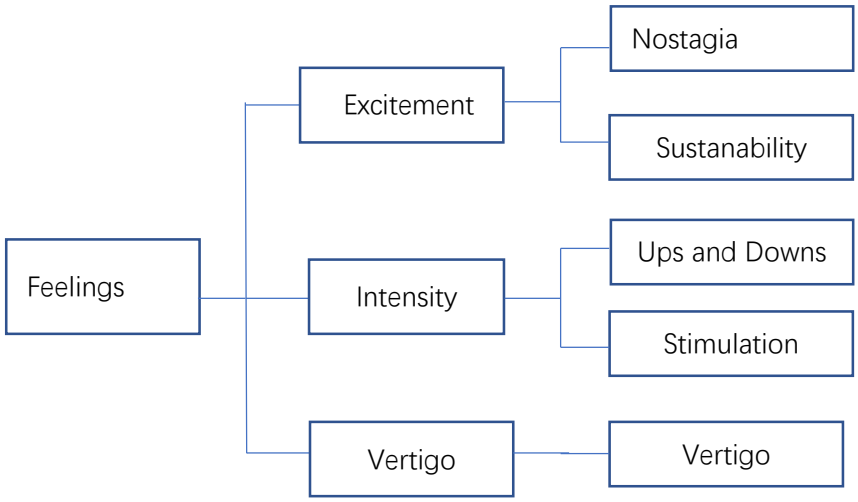


Figure 4 The Classification of Five Feeling Indexes

Our thinking figure of UI design is shown in Figure 5.

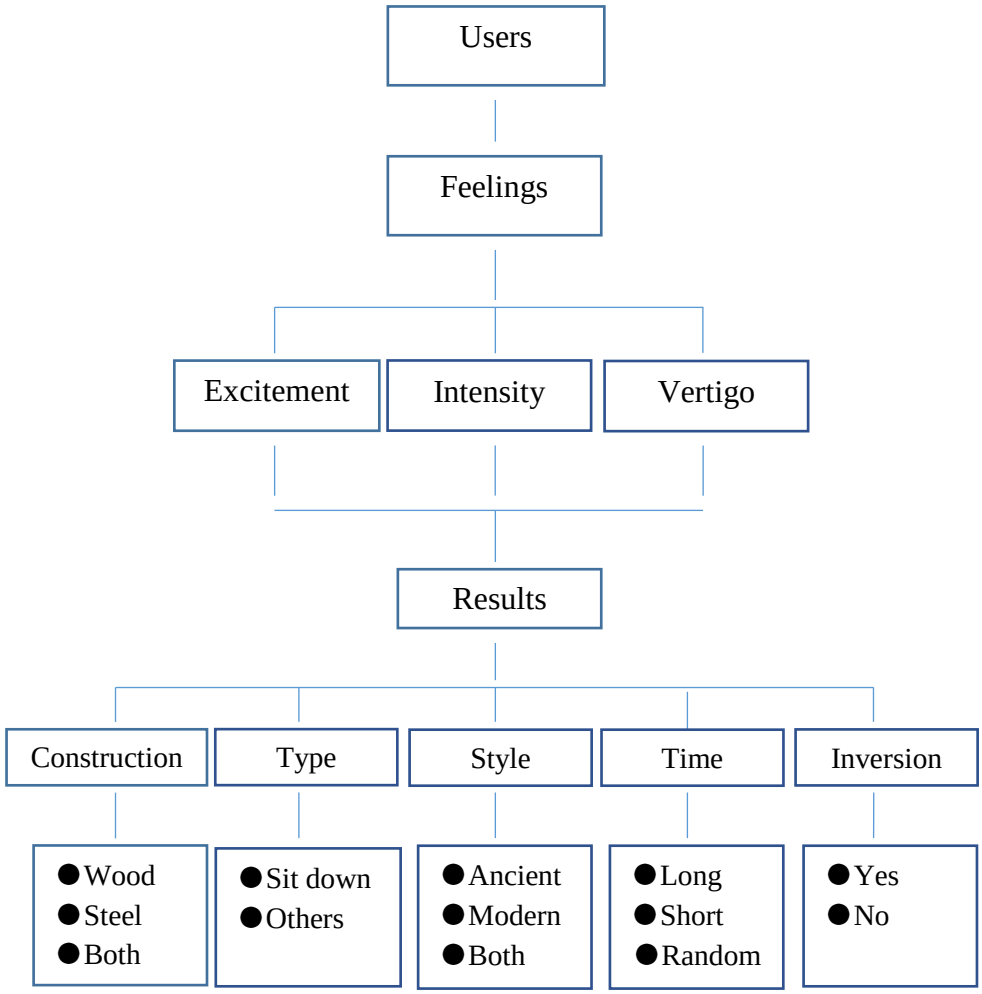


Figure 5 Option Design

7.2 Qualification for the Design

We define "User Friendly" as the clarity of the options, the numbers of options, and the speed a user can find the options. So, we follow the rules set by ISO (International Organization for Standardization) to determine the usability.

1.Learnability: How easy is it for users to accomplish basic tasks the first time they encounter the design?

2.Efficiency: Once users have learned the design, how quickly can they perform tasks?

3.Memorability: When users return to the design after a period of not using it, how easily can they re-establish proficiency?

4. Satisfaction: How pleasant is it to use the design?

We will follow the rules set above to qualify our design of app.

1.Learnability: Our app has only two major layers: a layer of feelings and a layer of construction and type. When the user sets his final preference, our app would provide an auto ranking, with almost no cost of learning.

2.Efficiency: Again, with only two major layers, the only thing to do to generate an auto ranking is two click of hand. Sometimes a user is not very sure what kind of feelings he wants to experience in the riding, so we also provide an overall choice, making them decide less and increasing their efficiency.

3.Memorability: Our structure of options is clear and involves many properties of roller coaster, so with a clear instruction, when a user see the app, he would instantly understand and re-establish proficiency.

4.Satisfaction: Though every user has his objective opinion of what is essential for a roller coaster to be satisfactory, our option design includes all the properties of roller coasters, which can satisfy almost every user with all the combinations possible.

7.3 UI Design of the App

We use a free UI designing web to create a simple UI design, as shown below in Figure 6.

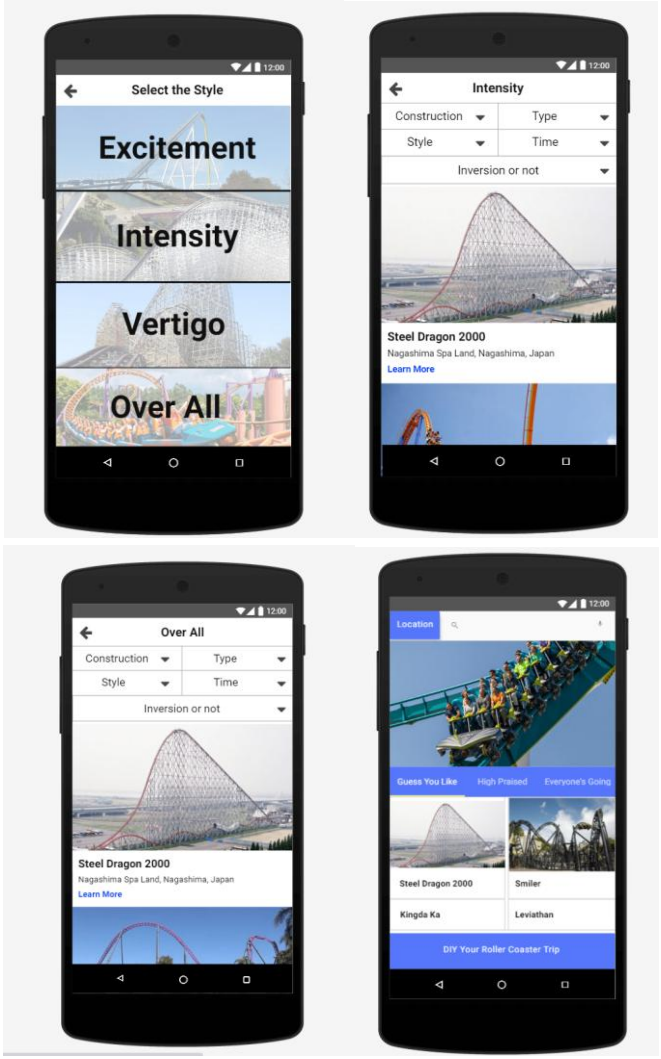


Figure 6 Graphs of UI Design

Note: all the graphs come from Internet search

Examples of rankings after choosing preference, see Figure7-Figure 9.

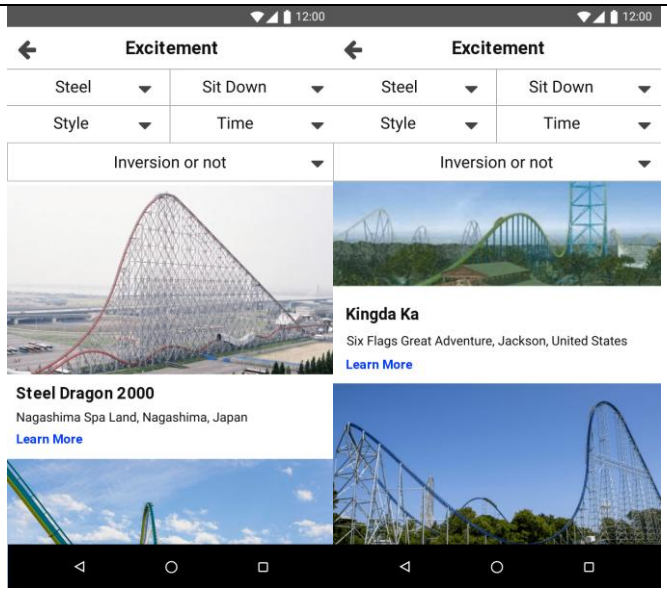


Figure 7 Excitement + Steel + Sit Down

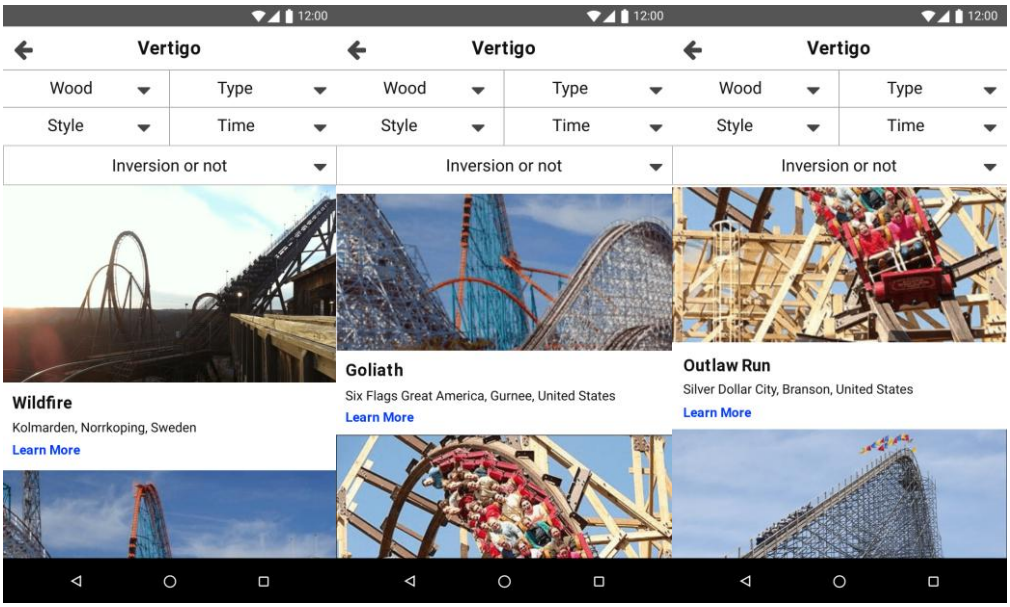


Figure 8 Vertigo + Wood + All Types

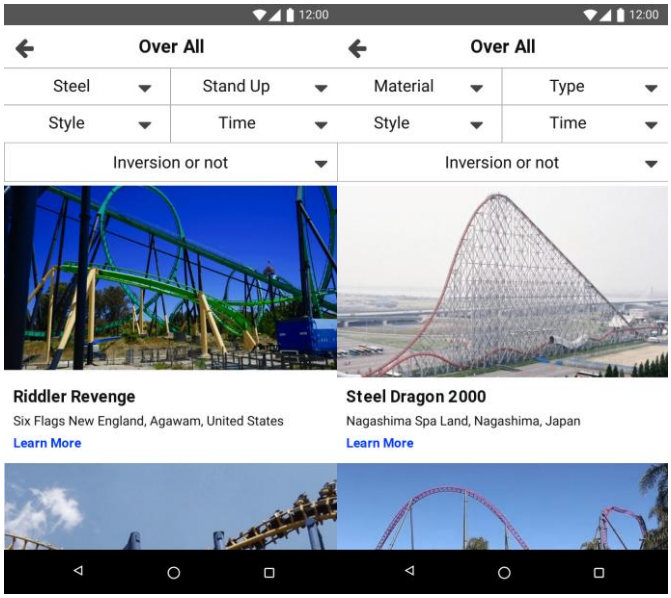


Figure 9 Overall + Steel + Stand Up

These three examples show how will the app generate the ranking based on personal preference.

8. Strength and Weakness of the model

8.1 Strength

When handling with the missing information, we didn't just delete them. Instead, we use cluster analysis to approximate the missing information. When handling with the missing information, we didn't just delete them. Instead, we use cluster analysis to approximate the missing information.

We use known data in the same category to estimate unknown data. Before clustering, we cluster according to different roller coaster characteristics of different constructions, which improves the accuracy of clustering, so the accuracy of missing data estimation is improved.

When establishing the evaluation model, we put forward five first-level indicators, among which there are several second-level indicators under the three indicators of ups and downs, sustainability and stimulation index. We use an entropy weight method to determine the weight coefficients of the second-level indicators. The weight coefficients determined by the entropy weight method are objective. These five indicators reflect the characteristics and performance of roller coaster from various aspects. Finally, a comprehensive evaluation index is established by preference coefficient. The preference coefficients in the model vary from person to person, which provides a feasible option for us to design apps to meet the individual needs of passengers.

8.2 Weakness

Although we have found a good way to supplement the missing data, there is a certain gap between the missing data and the real data, which will make the evaluation results inaccurate, especially in the G force data supplement. Because we build an algorithm through preference coefficients to facilitate the realization of passengers' personalized needs in apps, the ranking of 300 roller coasters is largely dependent on our preferences. Therefore, there will be great changes in results of sorting, when we focus on different angles, or the degree of preference for the same angle is different.

9. Conclusion

We built a function to assess the roller coaster with entropy method. When handling the incomplete information, we used cluster analysis and successfully filled the incomplete information. The clustering process is a fairly good cluster, with a cophenetic coefficient close to 1. The entropy method is an efficient way to determine whether an index is accurate and the contribution to the system. We also designed a app that perfectly fit the ISO standard of usability, with a visual UI on the phone. Of course, the model is far from perfect, with a positive coefficient for G force, since some people will have a comfortlessness under high g force condition.

10. Reference

[1]. https://www.ranker.com/crowdranked-list/best-roller-coasters?ref=lzyrltdlstszerg_rr
[2]. https://www.thetoptens.com/most-extreme-roller-coasters/

11.Appendix: The Ranking of All Roller Coaster

Rank	Name	Score	Rank	Name	Score
1	Steel Dragon 2000	5.05	34	Superman/ la Atracción de Acero	2.27
2	Smiler	4.00	35	Kraken	2.26
3	Kingda Ka	3.88	36	Bizarro	2.25
4	Leviathan	3.70	37	Intimidator	2.13
5	Fury 325	3.68	38	Katun	2.11
6	Millennium Force	3.50	39	Goliath	2.05
7	Top Thrill Dragster	3.26	40	Diamondback	1.92
8	Intimidator 305	3.25	41	Desperado	1.88
9	Fujiyama	3.05	42	Valravn	1.83
10	Steel Vengeance	2.96	43	Superman el Último Escape	1.83
11	Coaster Through the Clouds	2.95	44	10 Inversion Roller Coaster	1.81
12	Viper	2.95	45	Colossus	1.81
13	Alpengeist	2.91	46	Superman: Escape from Krypton	1.80
14	Dragon Mountain	2.89	47	Dragon's Run	1.79
15	Medusa	2.87	48	Vortex	1.78
16	Banshee	2.79	49	Monster	1.77
17	Formula Rossa	2.76	50	Takabisha	1.76
18	Titan	2.69	51	Pyrenees	1.73
19	Scream!	2.64	52	Mamba	1.68
20	Dragon Khan	2.61	53	Big One	1.68
21	Soaring Dragon & Dancing Phoenix	2.58	54	Steel Force	1.66
22	GateKeeper	2.55	55	Eejanaika	1.65
23	Silver Star	2.53	56	Incredible Hulk	1.64
24	Riddler's Revenge	2.52	57	Nitro	1.62
25	Montu	2.47	58	Superman the Ride	1.59
26	Kumba	2.44	59	Wild Thing	1.58
27	Hyperion	2.41	60	Tower of Terror II	1.53
28	Shambhala	2.38	61	Beast	1.52
29	Superman Krypton Coaster	2.34	62	Phaethon	1.52
30	Helix	2.33	63	Schwur des Kärnan	1.48
31	Altair	2.31	64	Raptor	1.43
32	Crazy Coaster	2.31	65	Flight of the Phoenix	1.42
33	Velikolukskiy Myasokombinat-2	2.31	66	Red Force	1.40

(Continued Table)

Rank	Name	Score	Rank	Name	Score
67	Flying Aces	1.39	104	Goliath	0.60
68	Afterburn	1.37	105	OCT Thrust SSC1000	0.59
69	Big Apple Coaster	1.36	106	Batman - The Dark Knight	0.57
70	Gao	1.35	107	Jupiter	0.53
71	Silver Bullet	1.35	108	Goliath	0.51
72	Behemoth	1.34	109	T Express	0.44
73	Hydra the Revenge	1.33	110	Talon	0.39
74	Flash	1.29	111	Incredicoaster	0.35
75	Cannibal	1.29	112	Phantom's Revenge	0.33
76	Hyper Coaster	1.27	113	iSpeed	0.31
77	Wildfire	1.26	114	Iron Rattler	0.28
78	Dinoconda	1.25	115	Fly the Great Nor'Easter	0.27
79	Happy Angel	1.23	116	Lightning Rod	0.27
80	Raging Bull	1.21	117	Batman The Ride	0.26
81	Rougarou	1.19	118	Batman The Ride	0.26
82	X2	1.18	119	Batman The Ride	0.26
83	Ride of Steel	1.12	120	Ultimate	0.25
84	Twisted Colossus	1.09	121	Batman The Ride	0.23
85	Superman - Ride Of Steel	1.09	122	Swarm	0.19
86	Voyage	0.99	123	Maverick	0.18
87	Griffon	0.96	124	El Toro	0.14
88	Goudurix	0.92	125	Goliath	0.11
89	Magnum XL-200	0.87	126	Stunt Fall	0.10
90	Tatsu	0.86	127	Blue Hawk	0.10
91	Thunder Dolphin	0.83	128	Storm Runner	0.09
92	blue fire Megacoaster	0.83	129	Extreme Rusher	0.07
93	Velikolukskiy Myasokombinat	0.82	130	Hades 360	0.05
94	Mako	0.82	131	Quimera	0.03
95	Do-Dodonpa	0.81	132	Batman The Ride	0.01
96	Apollo's Chariot	0.80	133	Wicked Cyclone	-0.01
97	Firehawk	0.75	134	Goliath	-0.04
98	Manta	0.73	135	Great White	-0.05
99	Patriot	0.69	136	Skyrush	-0.05
100	Batwing	0.68	137	Outlaw Run	-0.06
101	Fahrenheit	0.61	138	Batman the Ride	-0.06
102	Bullet Coaster	0.60	139	Desafio	-0.06
103	Soaring with Dragon	0.60	140	Firewhip	-0.06

(Continued Table)

Rank	Name	Score	Rank	Name	Score
141	Kong	-0.06	174	Full Throttle	-0.59
142	Limit	-0.06	175	GhostRider	-0.59
143	Mind Eraser	-0.06	176	New Revolution	-0.61
144	Mind Eraser	-0.06	177	Rock 'n' Roller Coaster	-0.62
145	MP-Xpress	-0.06	178	Time Traveler	-0.63
146	Raptor	-0.06	179	Stealth	-0.65
147	Riddler Revenge	-0.06	180	Flight Deck	-0.65
148	Flight of Fear	-0.06	181	Storm Chaser	-0.66
149	Batman: Arkham Asylum	-0.10	182	Steel Eel	-0.67
150	Flight Deck	-0.13	183	Twister II	-0.67
151	American Eagle	-0.15	184	Saw - The Ride	-0.70
152	Joker	-0.15	185	Demon	-0.71
153	Medusa Steel Coaster	-0.16	186	Demon	-0.73
154	Wodan Timbur Coaster	-0.16	187	Gemini	-0.74
155	Python in Bamboo Forest	-0.20	188	Montezum	-0.75
156	Shock Wave	-0.21	189	Flight of Fear	-0.78
157	Star Wars Hyperspace Mountain: Rebel Mission	-0.22	190	Temple of the Night Hawk	-0.80
158	New Texas Giant	-0.23	191	Montana Rusa	-0.83
159	Journey to Atlantis	-0.30	192	Colorado Adventure	-0.85
160	Boss	-0.31	193	Coaster Express	-0.86
161	Mr. Freeze Reverse Blast	-0.34	194	Big Thunder Mountain	-0.86
162	Mr. Freeze Reverse Blast	-0.34	195	Legend	-0.86
163	Big Loop	-0.40	196	Corkscrew	-0.89
164	Snow Mountain Flying Dragon	-0.40	197	Fluch von Novgorod	-0.91
165	Expedition GeForce	-0.43	198	Tonnerre de Zeus	-0.93
166	Superman - Ultimate Flight	-0.43	199	Invertigo	-0.93
167	Nemisis Inferno	-0.43	200	Steel Venom	-0.95
168	RailBlazer	-0.50	201	Corkscrew	-0.95
169	Taron	-0.51	202	Boomerang	-0.96
170	Black Mamba	-0.51	203	Boomerang	-0.96
171	Boulder Dash	-0.52	204	Boomerang	-0.96
172	Poltergeist	-0.53	205	Boomerang	-0.96
173	Wicked Twister	-0.58	206	Boomerang	-0.96

(Continued Table)

Rank	Name	Score	Rank	Name	Score
207	Boomerang	-0.96	237	Big Thunder Mountain Railroad	-1.43
208	Boomerang	-0.96	238	Sky Wheel	-1.45
209	Boomerang	-0.96	239	Mammut	-1.47
210	Boomerang	-0.96	240	Prowler	-1.48
211	Boomerang Coast to Coaster	-0.96	241	Mystic Timbers	-1.49
212	Flashback	-0.96	242	Wild One	-1.49
213	Apocalypse	-0.97	243	V2: Vertical Velocity	-1.53
214	Balder	-0.98	244	Jungle Trailblazer	-1.55
215	Grizzly	-1.00	245	Anaconda	-1.55
216	Doble Loop	-1.01	246	Alpina Blitz	-1.62
217	Abismo	-1.05	247	Bocaraca	-1.64
218	Eurosat Can Can Coaster	-1.07	248	Super Tornado	-1.64
219	Timber Drop	-1.08	249	Tornado	-1.64
220	Green Lantern Coaster	-1.10	250	Whirl Wind Looping Coaster	-1.64
221	Adrenaline Peak	-1.11	251	Terminator Salvation: The Coaster	-1.65
222	Nemesis	-1.17	252	Boardwalk Bullet	-1.66
223	Timber Wolf	-1.19	253	Sky Scream	-1.67
224	Vertical Velocity	-1.19	254	Furius Baco	-1.68
225	Journey to Atlantis	-1.21	255	Ranier Rush	-1.69
226	Joker	-1.27	256	Renegade	-1.71
227	Comet	-1.27	257	Montana Rusa	-1.71
228	Star Mountain	-1.28	258	Stampida	-1.74
229	Atlantica SuperSplash	-1.28	259	Giant Dipper	-1.75
230	Xcelerator	-1.28	260	Titan Cascabel	-1.77
231	Crazy Bird	-1.29	261	Timberhawk: Ride of Prey	-1.81
232	Texas Tornado	-1.30	262	Cedar Creek Mine Ride	-1.84
233	Apocalypse the Ride	-1.35	263	Viper	-1.85
234	Racer	-1.39	264	Mine Blower	-1.87
235	Nessie Superrollercoaster	-1.39	265	Montana Rusa	-1.88
236	Katapul	-1.41	266	Ravine Flyer II	-1.91

(Continued Table)

Rank	Name	Score
267	El Toro	-1.91
268	Half Pipe	-1.91
269	Iron Dragon	-1.93
270	Bandit	-1.93
271	Spatiale Experience	-1.97
272	Bat	-1.98
273	Montezooma's Revenge	-2.02
274	Cyclone	-2.03
275	Raven	-2.03
276	Oblivion	-2.04
277	Phoenix	-2.07
278	Ninja	-2.08
279	Screamer	-2.08
280	Coaster Thrill Ride	-2.12
281	Wild Thing	-2.17
282	SpeedSnake FREE	-2.21
283	Whizzer	-2.27
284	Giant Dipper	-2.30
285	Piraten	-2.34
286	Revenge of the Mummy the Ride	-2.34
287	Kawazemi	-2.35
288	Road Runner Express	-2.36
289	Blue Streak	-2.49
290	Pandemonium	-2.53
291	Desert Race	-2.54
292	Judge Roy Scream	-2.61
293	Sidewinder	-2.68
294	HeiBe Fahrt	-2.73
295	Manta	-2.74
296	Taunusblitz	-2.78
297	Force One	-2.85
298	Winjas	-2.95
299	Pandemonium	-3.51
300	Backlot Stunt Coaster	-3.59